

EE 4113 – ECE Lab 2
Spring 2026
Lab #01 Report

Student 1 Name: Ivan Garcia
UTSA ID: eaa108
Section: 0B1
Group Number: 19

Student 2 Name: Tristan Herring
UTSA ID: xaw089
Section: 0B1
Group Number: 19

Objective:

Oscilloscopes are used to measure voltage signals. Passive oscilloscope probes measure the voltage difference between a test point within the circuit and the circuit's ground point. Due to differences in the voltage signal being measured and the probe scope settings, (i) the oscilloscope settings will need to match the probe settings, and (ii) the probe will need to be calibrated (referred to as compensating the probe). Operating the oscilloscope and the probe will not obtain accurate measurements if these two preliminary checks are not completed. Poorly compensated probes can lead to measure inaccuracies such as incorrect amplitudes and distorted waveforms. These inaccuracies increase as the frequency increases. In this exercise we will look at how to compensate the oscilloscope, along with different considerations to make when using the oscilloscope to measure signals that come from a function generator. This lab will give you the skills and experience utilizing the oscilloscope and function generator.

Equipment List:

Table 1 Equipment List

<i>Description</i>	<i>Manufacturer</i>	<i>Model Number</i>
Solderless Breadboard	R.S.R. Electronics	MB108
Function Generator	Lab-Volt	A8771
Oscilloscope and Probes	Tektronics	DM2510
Power Supply	BK Precision	1762

Parts List:

Table 2 Parts List

<i>Quantity</i>	<i>Part</i>	<i>Value</i>
1	Resistor	50 ohm ($\frac{1}{4}$ Watt)

Procedure:

Part A: Compensating the Probe: Over-compensation vs. Under-compensation

- The oscilloscope is powered on using the front-panel power switch, reset to a known configuration using the Default Setup button, and a passive probe is connected to the Channel 1 input using the BNC connector and locking ring, after which the probe attenuation switch is set to 10 $\times$ and the channel menu is checked to verify that the oscilloscope is configured for a 10 $\times$ probe.
- The probe ground lead is connected to the ground terminal of the Probe Comp output and the probe tip is connected to the signal terminal, after which the Autoset function is used to display the square-wave reference signal on the oscilloscope screen.
- The probe compensation adjustment screw is slowly rotated using the adjustment tool until the displayed square wave exhibits overshoot, indicating an overcompensated condition, at which point a peak-to-peak voltage measurement is added to the display if necessary and the waveform image is saved to a USB storage device.

- The function generator is powered on and configured to output a 3 V_{pp}, 80 kHz sine wave with 0 V DC offset, the probe ground lead is connected to the generator ground and the probe tip is connected to the output signal, the generator output is enabled, the oscilloscope Autoset function is applied, the peak-to-peak voltage is recorded, and the waveform image is saved before disabling the generator output.
- The probe is reconnected to the Probe Comp terminals and autoset is applied, after which the probe compensation adjustment screw is rotated until the square wave displays a rounded, overdamped response characteristic of an undercompensated condition, and the resulting waveform image is saved to a USB device.
- The probe is then connected to the function generator output, the generator output is enabled, autoset is applied on the oscilloscope, the peak-to-peak voltage is recorded, and the waveform image is saved before disabling the generator output.
- The probe is once again reconnected to the Probe Comp terminals and autoset is applied, after which the compensation adjustment screw is carefully tuned until the square wave exhibits flat, well-defined, squared-off edges indicating proper probe compensation, and the waveform image is saved to a USB device.
- The probe is connected to the function generator output, the generator output is enabled, autoset is applied, the peak-to-peak voltage is recorded, and the waveform image is saved before disabling the generator output.
- Using the measured peak-to-peak voltage and assuming a 10 M Ω probe resistance, the internal peak-to-peak voltage of the function generator is calculated using a voltage divider model, and the accuracy and waveform quality of the overcompensated, undercompensated, and properly compensated measurements are compared using the recorded oscilloscope images.

Part B: Direct Function Generator Measurements with the Oscilloscope vs. Measurements Taken Across a Circuit Load

- A BNC component with a 50 Ω resistor soldered across its terminals is connected to the output of the function generator, with care taken to avoid stressing or breaking the resistor leads at the solder joints, and the oscilloscope ground lead is attached to the side of the resistor that is soldered to the BNC encasement while the oscilloscope probe tip is attached to the side of the resistor that is soldered to the signal line of the BNC connector.
- If the BNC component with an integrated 50 Ω resistor is unavailable, an approximately 50 Ω resistor (such as 56 Ω) is placed into a breadboard to serve as a load impedance, the red and black leads from the function generator are connected across the resistor, the oscilloscope ground lead is connected to the same side of the resistor as the black lead, and the oscilloscope probe tip is connected to the same side of the resistor as the red lead.
- The function generator output is enabled, the Autoset function on the oscilloscope is used to properly display the waveform, the peak-to-peak voltage measurement is recorded or added to the display if necessary using the measurement menu, and the waveform image is saved to a USB device as a representation of the function generator signal applied across a 50 Ω load.
- Using the recorded peak-to-peak voltage and assuming a 50 Ω load resistance, the internal peak-to-peak voltage of the function generator is calculated using a voltage divider model while neglecting probe resistance, and the calculated value is compared to the peak-to-peak voltage indicated on the function generator display as well as to previous results obtained without a load.

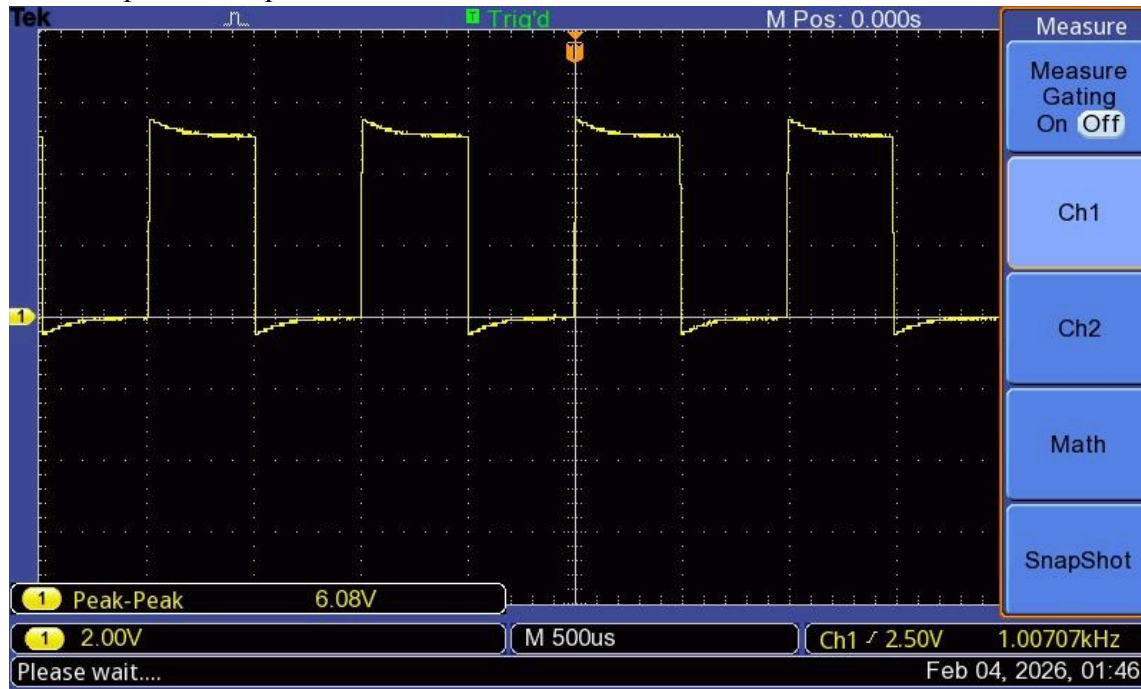
- The relationship between the internal function generator voltage and the voltage shown on the function generator display is discussed in the lab report based on the calculated and measured values.

- After completing all measurements, the circuit is disassembled, all cables and components are packed away, the workstation is cleared, and laboratory verification and equipment return procedures are completed.

Results:

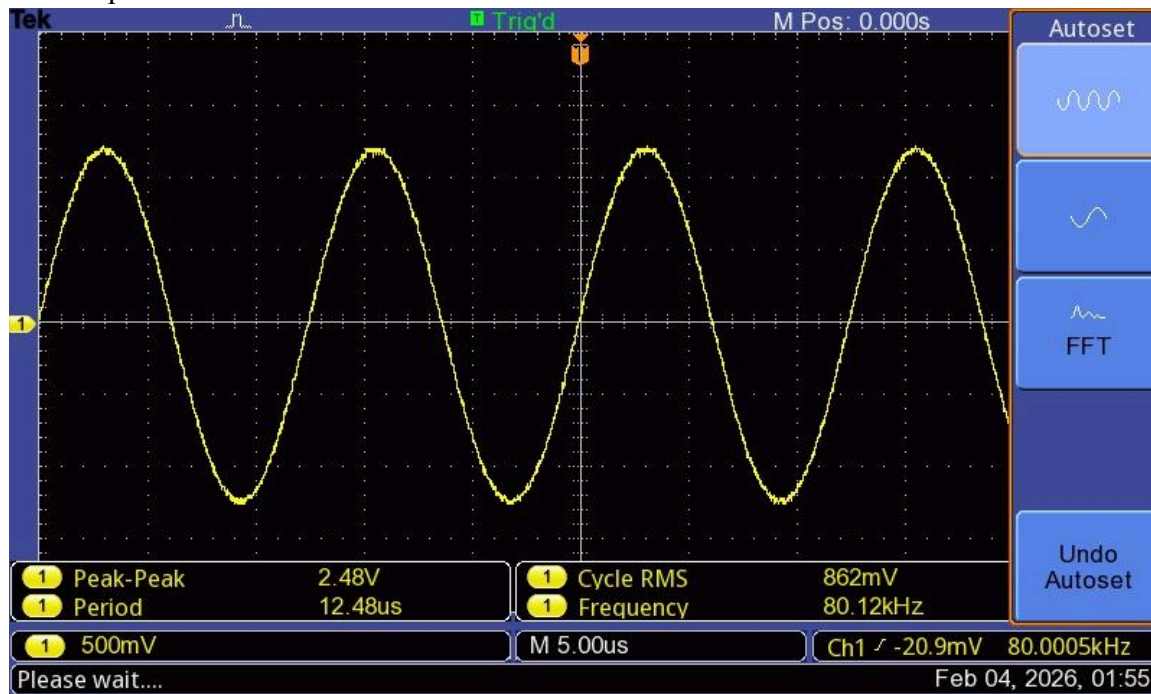
Part A:

Overcompensated squarewave:

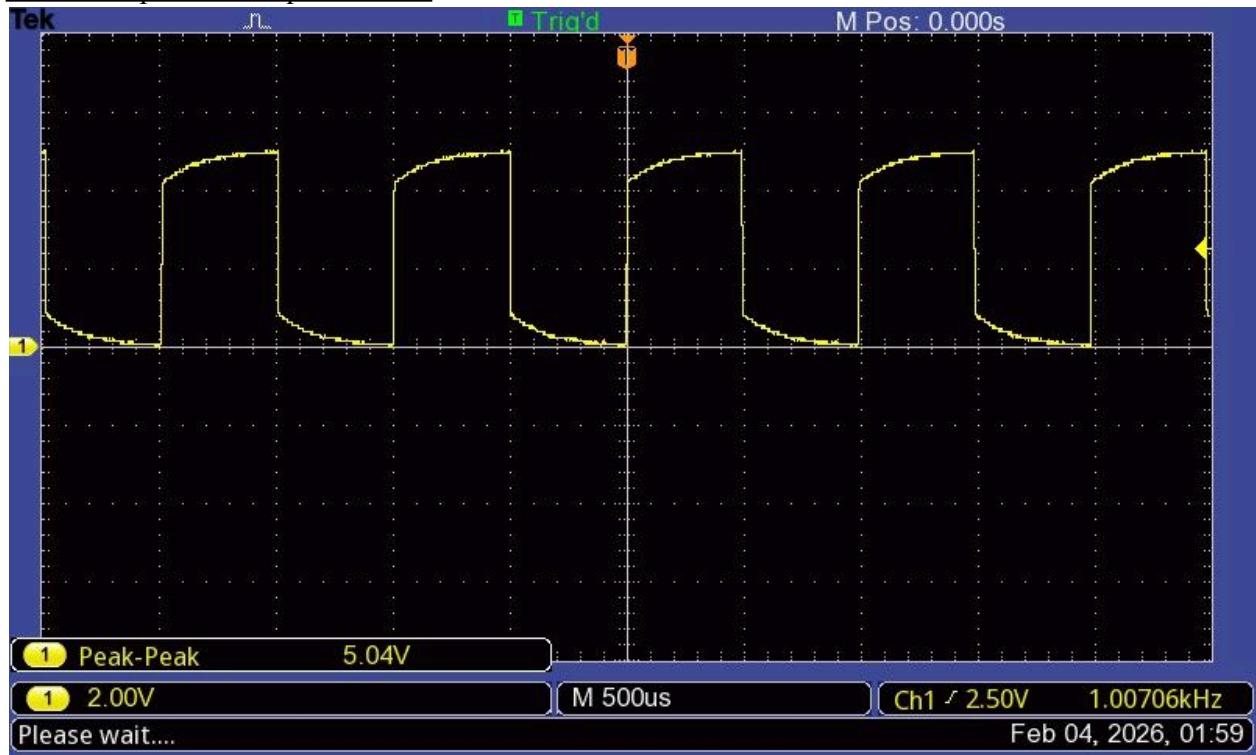


Overcompensated Sinewave:

Peak to peak is: 2.48V

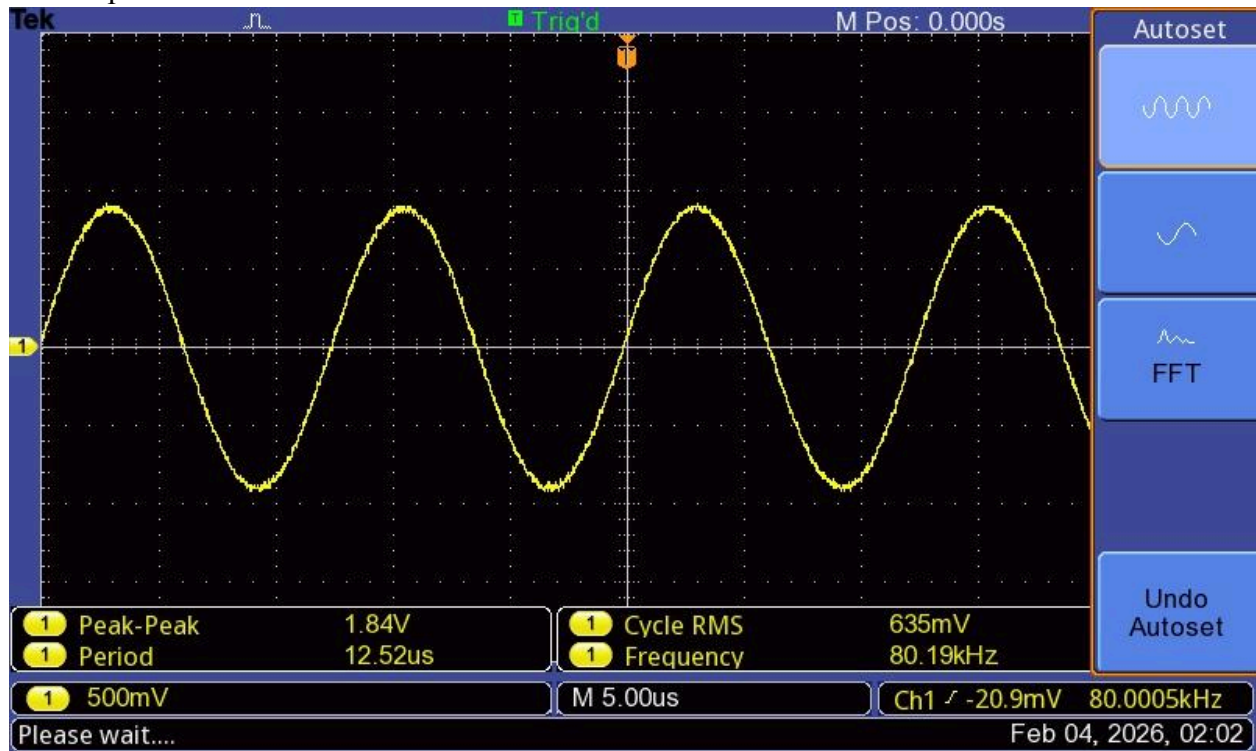


Undercompensated Square wave:

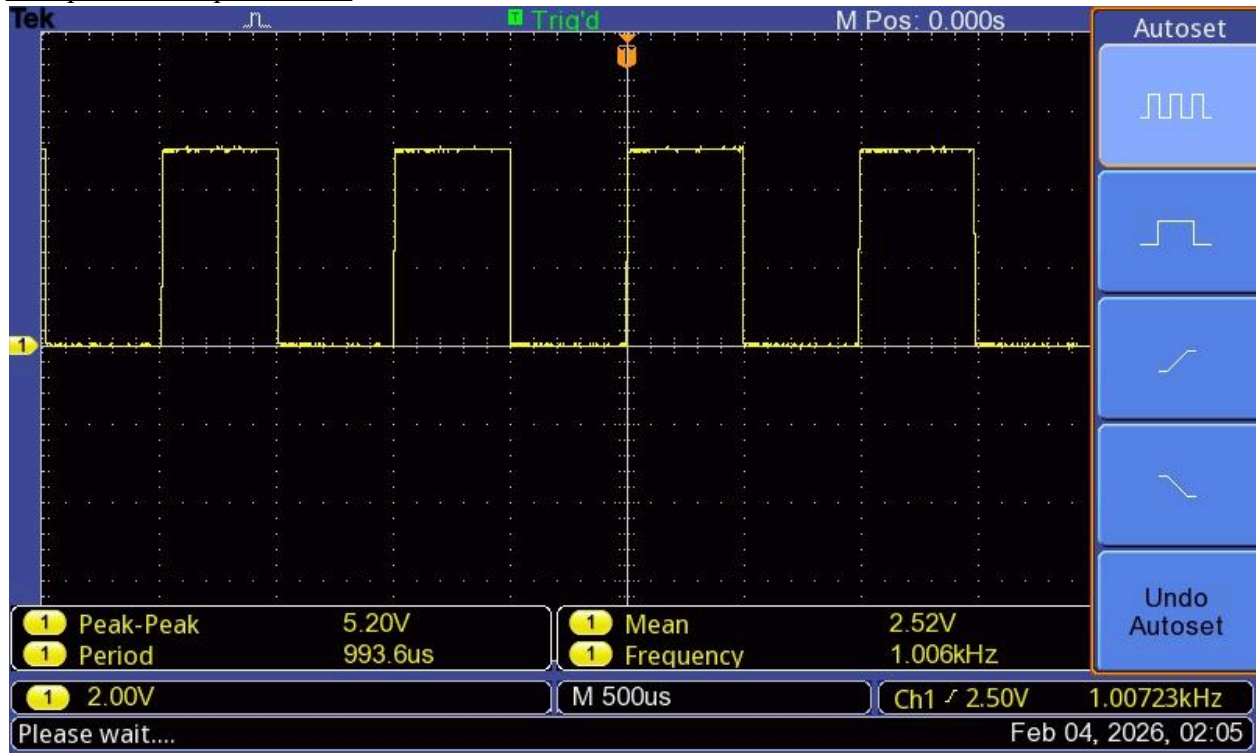


Undercompensated Sinewave:

Peak to peak: 1.84V

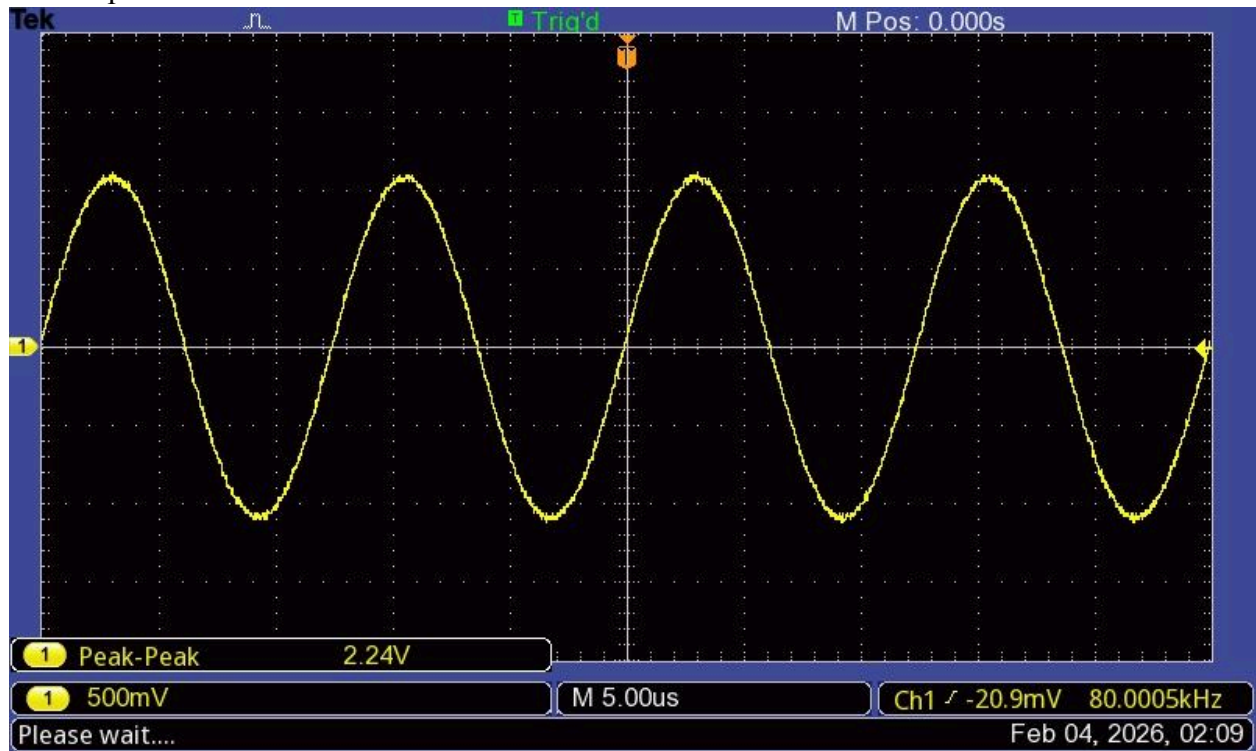


Compensated Squarewave:



Compensated Sinewave:

Peak to peak: 2.24V

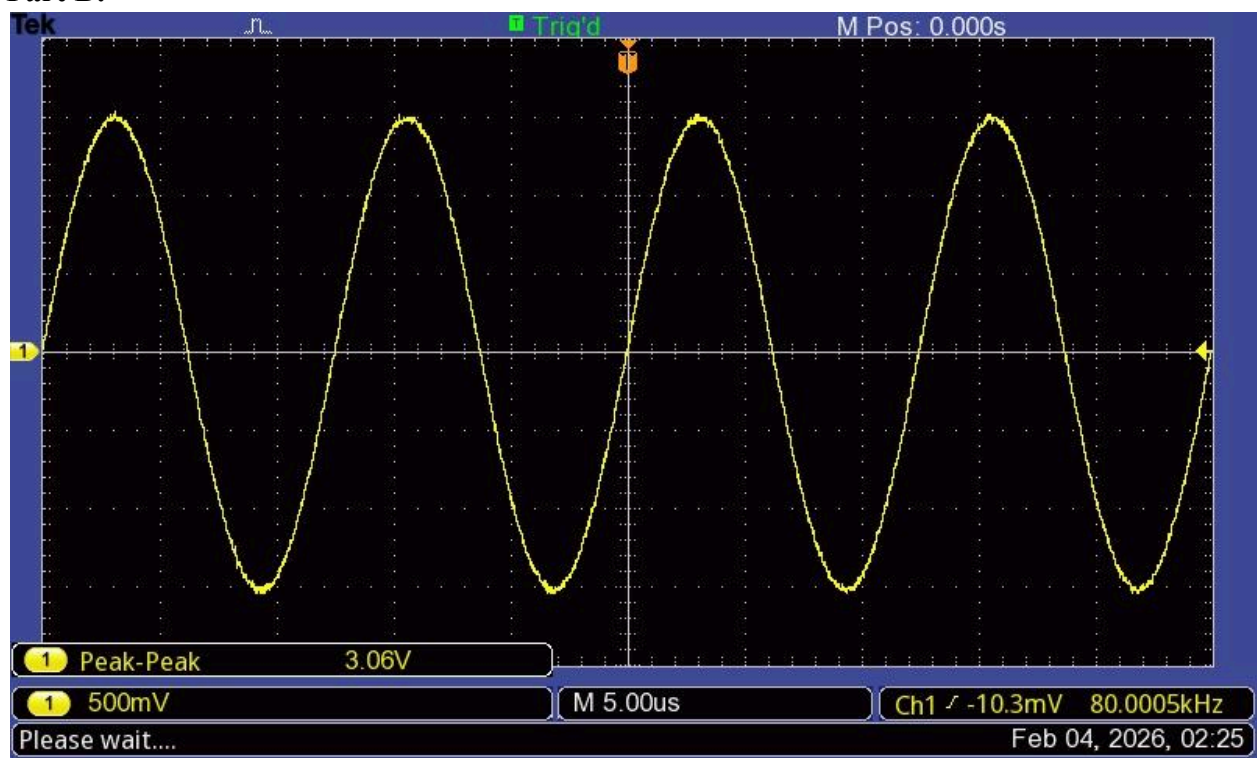


$$V_S = (R_S + R_L)V_L/R_L$$

$$(50+10M)2.24/10M = 2.24V = \text{peak to peak voltage of function generator}$$

The calculated and measured voltages are matching with only a slight difference in the 10^{-5} position. Thus, the two peak to peak voltages are approximately the same and are directly related.

Part B:



Peak to peak voltage is 3.06

Using the equation $V_S = (R_S + R_L)V_L/R_L$ we can plug in values to find V_S

$$(50)3.06/50 = 3.06V = \text{peak to peak voltage of function generator}$$

In this situation the relationship between the measured and displayed P-P voltages are directly related, and still equal. The calculated voltage is the same as the display voltage. This is the same relationship as what was found in A, except for the values being more equal, with no approximations.

Conclusion:

Differences in sine waves:

The overcompensated sine wave had too high of a peak to peak voltage, while the undercompensated sine wave had too low a voltage. Using the formula of $|(V_A - V_E)/V_E| * 100$ we can get the percent error for each value. The overcompensated percent error is 10.7% while the undercompensated percent error is 17.86%. Both of these values are fairly high with the undercompensated value being well beyond allowable error. Thus, we know for sure that the compensation of the sine waves via oscilloscope probes are a vital piece in lab testing, because lack of compensation will result in huge errors.

Function generator values:

Based on the relationships found in the ends of parts A and B, we can firmly say that the measured and shown voltage p-p values are approximately the same, regardless of the probe resistances. Which is to say, we know that the device is functioning normally and will provide accurate voltage values.